



Neck level algorithmic approach to trauma

Mehr Qureshi, MD, FACS, Krishan Patel, MD, FACS, Ronald Simon, MD, FACS

From the Division of Acute Care Surgery, Department of Surgery, Maimonides Medical Center, Brooklyn, New York



KEYWORDS

Neck zones;
Blunt trauma;
Penetrating trauma;
Algorithm;
Management;
Neck trauma;
Neck injury

Trauma to the neck poses a unique set of challenges owing to the presence of multiple important structures lying in close proximity to each other. A thorough understanding of the anatomy of the neck as well as having a step wise approach for timely evaluation and management, are crucial in identifying and managing neck injuries. Trauma to the neck can occur due to blunt or penetrating mechanisms. Though the injury patterns may differ, the initial management is similar and follows an algorithm based approach based on anatomic landmarks as well as patient stability. Various imaging and diagnostic modalities are available and are utilized in the prompt diagnosis of cerebrovascular and aerodigestive neck injuries.

© 2020 Elsevier Inc. All rights reserved.

Introduction

When considering the management of neck trauma, the most critical factor is whether the mechanism is blunt or penetrating. Blunt trauma comprises the majority of neck injuries and includes motor vehicle collisions, falls, and sport injuries; hanging and strangulation comprise a special subset. The cervical spine and the laryngotracheal airway are most commonly injured in blunt trauma.¹ Additionally, blunt cerebrovascular injury (BCVI) may occur and can be life threatening. Penetrating trauma represents 5%-10% of all neck trauma and describes injury that has breached the platysma muscle.² Stab and gunshot wounds constitute the majority of such injuries, with gunshot wounds causing more significant injury to a larger number of structures in the neck and overall requiring more frequent operative intervention.³

Penetrating trauma to the neck can cause injuries to multiple organ systems: Vascular (occlusion, dissection, extravasation, or fistula formation); aerodigestive (which are associated with high morbidity and mortality especially if diagnosis and treatment are delayed); neurologic (from injuries to the spinal cord, brachial plexus, and cranial nerves VII-XII); and injury to the upper thorax (leading to hemo- or pneumothoraces).²

Anatomy

Owing to complex anatomy, the neck has classically been divided into 3 zones extending from caudal to cranial, with additional anterior and lateral divisions (Figure 1). Zone I is the most caudal zone and extends from the level of the sternal notch/clavicles to the cricoid cartilage. Zone II continues from the cricoid cartilage to the angle of the mandible. Zone III includes the angle of the mandible to the base of the skull. Additionally, the posterior border of the sternocleidomastoid muscle divides the neck into anterior and posterior sections. Table 1 depicts the zones of the neck and the various structures found within each

Address reprint requests and correspondence: Ronald Simon, MD, Division of Acute Care Surgery, Department of Surgery, Maimonides Medical Center, 4901-A Fort Hamilton Parkway, Brooklyn, NY 11219.

E-mail address: rosimon@maimonidesmed.org

<http://doi.org/10.1016/j.otot.2020.10.005>

1043-1810/© 2020 Elsevier Inc. All rights reserved.